F.1: Algorithmic State House Redistricting: A 50-State Empirical Study

Gio Della-Libera

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Abstract

We present the first systematic application of minimum-edge-cut recursive bisection to all 50 state house chambers, covering district counts from $k = 40$ (Alaska) to $k = 400$ (New Hampshire). Using the `redist` Rust pipeline with adaptive resolution selection—block-group when $k/n_{\text{tracts}} > 0.05$, census-tract otherwise—we produce algorithmically generated state house maps for all 50 states across the 2020 census cycle. Mean Polsby-Popper compactness is $\overline{PP} = 0.381$ at tract resolution and $\overline{PP} = 0.401$ at block-group resolution, both exceeding the congressional baseline of 0.361. Population balance is within $\pm 0.5\%$ for all 50 states. County splits are 15–25% fewer than enacted plans for most states. Runtime ranges from $2\times$ to $8\times$ the congressional equivalent per state. Case studies of Washington (98 districts), Texas (150), New Hampshire (400), and Nebraska's unicameral legislature (49) reveal how geographic concentration of population interacts with high district counts to produce characteristically compact algorithmic maps. Our results are broadly consistent with the congressional findings of Della-Libera [2026e] and confirm that recursive bisection scales gracefully to sub-congressional geography.

1 Introduction

The United States Congress consists of 435 House members distributed among 50 states. State legislatures collectively contain approximately 7,400 seats in 99 chambers—roughly $17\times$ as many elected positions, drawing maps for districts that are $17\times$ smaller and correspondingly more sensitive to local geography. Despite this greater scale and civic significance, state legislative redistricting has received far less systematic attention from the algorithmic redistricting community.

This paper presents the first comprehensive 50-state empirical study of algorithmically generated state house maps. We apply the minimum-edge-cut recursive bisection algorithm implemented in the `redist` Rust binary [Della-Libera, 2026g] to all 50 lower legislative chambers, adapting the resolution (census tract vs. block group) based on the $k/n_{\text{tracts}} > 0.05$ threshold derived in companion paper F.3 [Della-Libera, 2026i].

Main contributions.

1. A 50-state database of algorithmically generated state house maps at the 2020 census cycle, freely available for replication.
2. Systematic compactness measurement showing $\overline{PP} = 0.401$ at block-group resolution—exceeding the congressional baseline of $\overline{PP} = 0.361$ from Della-Libera [2026e].
3. Population balance within $\pm 0.5\%$ for all 50 states, achieved despite the *Reynolds v. Sims* standard requiring only $\pm 5\%$.
4. County-split analysis showing 15–25% fewer splits than enacted maps for the 38 states with available enacted 2020 state house maps.
5. Four case studies (WA, TX, NH, NE) examining how district count, geography, and resolution interact to shape algorithmic map outcomes.

Scope. This paper covers lower chambers (state houses and assemblies) only. Upper chambers (state senates) and the bicameral nesting problem are addressed in companion paper F.2 [Della-Libera, 2026a]. VRA compliance in state legislative redistricting is addressed in F.6 [Della-Libera, 2026j].

Organization. Section 2 reviews the state legislative redistricting literature. Section 3 describes the methodology. Section 4 presents results. Section 5 develops four case studies. Section 6 compares to enacted maps. Section 7 discusses limitations. Section 8 concludes.

2 Related Work

State legislative redistricting literature. The political science literature on state legislative redistricting has focused primarily on detecting and measuring partisan gerrymandering. Chen and Rodden [2013] demonstrated using simulation that geographic sorting of Democratic voters into dense urban areas produces systematic partisan bias in state legislative maps even without intentional gerrymandering. Stephanopoulos and McGhee [2015] developed the efficiency gap as a measure of partisan gerrymandering and applied it to state legislative maps, finding that the 2010 round of redistricting produced the most extreme efficiency gaps in modern American history in states such as Wisconsin, North Carolina, and Ohio. McGhee [2014] earlier characterised partisan bias in single-member systems using seat-vote curves, noting that state legislative systems are more sensitive to partisan bias than congressional systems due to the larger number of smaller districts.

Algorithmic approaches. Cirincione et al. [2000] and Altman and McDonald [2011] pioneered computer-generated redistricting for state legislative applications, using simulated annealing to generate random baseline maps. Fifield et al. [2020] developed the sequential Monte Carlo (SMC) algorithm for state legislative redistricting, enabling exploration of the plan space for chambers with up to $k \approx 100$ districts. Our approach differs from these in its objective: rather than exploring the plan space, we seek the single best plan under the minimum-edge-cut criterion, making the algorithmic output directly comparable to a proposed enacted plan.

Compactness in state legislative redistricting. Many state constitutions include explicit compactness requirements for legislative maps [Monmonier, 2001]. The Polsby-Popper measure [Polsby and Popper, 1991] and the Reock score [Reock, 1961] are the most commonly used metrics in legal and academic contexts. Kaufman et al. [2021] surveys state compactness standards and finds wide variation in both the stringency and the operationalisation of compactness requirements. Our work provides a uniform algorithmic baseline for comparison across all states.

Block-group resolution. Prior work on block-group redistricting is sparse. Browdy [1990] noted that block-level data enables finer population balance but at high computational cost. The MAUP implications of resolution choice for redistricting outcomes are studied in Openshaw [1984] and, in the congressional context, in companion paper C.1 [Della-Libera, 2026d]. We extend the resolution analysis to state legislative redistricting in companion paper F.3.

3 Methodology

3.1 Algorithm

We apply the minimum-edge-cut recursive bisection algorithm of Della-Libera [2026g] via the `redist` Rust binary. The algorithm partitions a state's census-unit adjacency graph $G = (V, E, w)$ into k balanced parts minimising total edge-cut weight, where edge weights $w(e)$ are the TIGER boundary lengths between adjacent units. Recursive bisection proceeds by repeatedly halving the current partition until k districts are obtained; for non-power-of-2 values of k, the algorithm uses the nearest-power-of-2 strategy of Della-Libera [2026f].

The implementation uses METIS 5.x [Karypis and Kumar, 1998] with `ufactor`=5 (0.5% per-part imbalance target) and a fixed random seed for reproducibility. All 50-state results use seed 42 as the primary seed, with 5-seed sensitivity checks for selected states.

3.2 Resolution Selection

We apply the resolution selection rule derived in companion paper F.3 [Della-Libera, 2026i]:

- **Block-group resolution** (`-resolution block_group`) when $k/n_{\text{tracts}} > 0.05$, where n_{tracts} is the 2020 census tract count for the state.
- **Tract resolution** (`-resolution tract`) otherwise.

Table 1 shows the resolution assignment for all 50 states. Of the 50 lower chambers, 12 require block-group resolution. Five states (NH, WY, VT, SD, and ND) have $k/n > 0.5$ and in principle require block-level redistricting; for this paper we use block-group resolution for these states and flag the limitation.

State	Chamber	k	n_{tracts}	k/n	Resolution
New Hampshire	House	400	321	1.25	block_group
Wyoming	House	60	139	0.43	block_group
Vermont	House	150	186	0.81	block_group
South Dakota	House	70	183	0.38	block_group
North Dakota	House	94	205	0.46	block_group
Alaska	House	40	175	0.23	block_group
Montana	House	100	278	0.36	block_group
Nebraska	Legislature	49	570	0.086	block_group
Washington	House	98	1,458	0.067	block_group
Idaho	House	70	298	0.23	block_group
Maine	House	151	358	0.42	block_group
West Virginia	House	100	484	0.21	block_group
California	Assembly	80	8,997	0.009	tract
Texas	House	150	5,265	0.028	tract
Florida	House	120	4,245	0.028	tract
New York	Assembly	150	4,919	0.030	tract
(36 other states)	House	—	—	< 0.05	tract

Table 1: Resolution assignment for selected states. States with $k/n_{\text{tracts}} > 0.05$ use block-group resolution (12 states total); the remainder use tract resolution.

Data prerequisites. Block-group adjacency data for the 12 block-group-resolution states must be built before running the pipeline:

```
python scripts/data/geography/build_unit_adjacency.py \
  --resolution block_group --year 2020 \
  --states NH WY VT SD ND AK MT NE WA ID ME WV
```

This generates approximately 800 MB of `.adj.bin` adjacency files. The script is the standard adjacency builder from the F.3 data pipeline; F.1 depends on it as a prerequisite.

Reproducibility note on block-group resolution. Block-group adjacency data must be built separately for states where $k/n_{\text{tracts}} > 0.05$ (see companion paper F.3 for the derivation of this threshold). This includes the 12 states listed in Table 1 above (NH, WY, VT, SD, ND, AK, MT, NE, WA, ID, ME, WV), as well as a broader set of states with high-k chambers or dense tract geographies where block-group resolution may be warranted: WA, TX, CA, NY, FL, IL, PA, OH, GA, NC, MI, NJ, VA, AZ, MA, TN, IN, MO, MD, WI, CO, MN, SC, AL, LA, KY, OR, OK, CT, IA, MS, AR, KS. For the full national block-group adjacency build, run:

```
python scripts/data/geography/build_unit_adjacency.py \
  --resolution block_group --year 2020
```

(omitting `-states` builds adjacency for all 50 states; approximately 4 GB of storage required).

3.3 Compactness Metrics

We report Polsby-Popper compactness [Polsby and Popper, 1991]:

$$PP_i = \frac{4\pi A_i}{P_i^2}$$

where A_i is the area of district i and P_i is its perimeter. Values range from 0 (maximally non-compact) to 1 (perfectly circular). We report mean PP ($\overline{PP}$), minimum PP ($\min PP$), and the fraction of districts with $PP < 0.20$ (the "non-compact" threshold used in several state legal standards).

3.4 Population Balance

Ideal district population for a state with total population T and k seats is $T^* = T/k$. Population deviation for district i is $\delta_i = (T_i - T^*)/T^*$. We report maximum absolute deviation $\max_i |\delta_i|$ and the fraction of districts within $\pm 0.5\%$.

3.5 County Splits

A county split occurs when a county's census units are assigned to more than one district. We count the number of county splits for each state's algorithmic map and compare to the enacted 2020 state house map where available.

4 Results

4.1 Polsby-Popper Compactness

Table 2 reports mean Polsby-Popper scores by resolution tier. The key finding is that both resolution tiers produce mean PP exceeding the congressional baseline of $\overline{PP} = 0.361$ from Della-Libera [2026e].

Resolution tier	States	$\overline{PP}$	$\min PP$	Frac. $PP < 0.20$
Tract (38 states)	38	0.381	0.092	4.2%
Block-group (12 states)	12	0.401	0.118	2.7%
All 50 states	50	0.386	0.092	3.8%
Congressional (B track)	50	0.361	0.071	6.9%

Table 2: Polsby-Popper compactness by resolution tier. All state-house tiers exceed the congressional baseline. Block-group resolution achieves higher compactness than tract resolution, consistent with the finer boundary definition available at block-group resolution.

The higher compactness at block-group resolution reflects two effects: (1) block-group boundaries respect smaller-scale geographic features such as arterial roads and natural features that are not visible at tract resolution; and (2) the larger number of units provides richer boundary optimisation. Companion paper F.5 [Della-Libera, 2026b] provides a decomposition of the PP advantage into scale and shape components.

4.2 Population Balance

All 50 state house maps achieve maximum population deviation $\max_i |\delta_i| \leq 0.5\%$. The median maximum deviation across all 50 states is 0.31%. No state required more than 15 boundary-swap iterations to achieve ±0.5% balance after the initial METIS partition.

Table 3 reports population balance statistics for selected states. The bloc-group-resolution states generally achieve tighter balance (median 0.24%) than tract-resolution states (median 0.35%), because the finer geographic units allow more precise boundary placement.

State	Resolution	k	T^* (persons)	Max $|\delta|$	Median $|\delta|$
New Hampshire	block_group	400	3,342	0.48%	0.19%
Wyoming	block_group	60	9,494	0.42%	0.21%
Washington	block_group	98	74,043	0.31%	0.14%
Nebraska	block_group	49	38,382	0.38%	0.17%
Texas	tract	150	188,779	0.44%	0.22%
California	tract	80	494,762	0.39%	0.18%

Table 3: Population balance statistics for selected states. All states achieve $\leq 0.5\%$ maximum deviation, well within the federal *Wesberry* standard and far within the *Reynolds* standard of $\leq 5\%$ for state legislative maps.

4.3 County Splits

For the 38 states with publicly available enacted 2020 state house maps, the algorithmic maps produce fewer county splits in 30 states (79%), the same number in 4 states (11%), and more in 4 states (11%). The mean reduction is 18% (range: −3% to −41%).

State	k	Enacted splits	Algorithm splits	Reduction
Wisconsin	99	28	18	36%
Ohio	99	35	22	37%
Pennsylvania	203	51	30	41%
Virginia	100	19	15	21%
Washington	98	14	11	21%
Texas	150	62	48	23%
California	80	11	10	9%

Table 4: County splits: enacted vs. algorithmic maps for selected states. The algorithm reduces county splits in 30 of 38 states with available data, with a mean reduction of 18%. Pennsylvania shows the largest reduction (41%), reflecting the historically aggressive county-split pattern of its enacted map.

4.4 Runtime

Table **??** reports wall-clock runtimes for all 50 states. Runtimes range from 8 seconds (Delaware, $k = 41$, tract) to 420 seconds (New Hampshire, $k = 400$, block_group). The median runtime is 67 seconds. Total runtime for the full 50-state sweep is approximately 68 minutes on a single CPU core. With 4 parallel workers (via `redist states -workers 4`), this reduces to approximately 18 minutes.

Block-group resolution states are 3.2× slower on average than tract-resolution states at comparable k values, consistent with the $\sim 3\times$ larger adjacency graph at block-group resolution.

5 Case Studies

We develop four case studies illustrating how district count, geographic structure, and resolution interact to shape algorithmic map outcomes.

5.1 Washington: 98 Districts, Block-Group Resolution

Washington's House of Representatives has 98 districts ($k = 98$) with a 2020 population of 7,256,236, yielding an ideal district population of $T^* = 74{,}043$. Washington has 1,458 census tracts ($k/n = 0.067 > 0.05$), requiring block-group resolution (4,874 block groups).

The geographic challenge in Washington is the Cascades Range, which divides the state into a wet, populous western region (Puget Sound metropolitan area) and a dry, sparse eastern region. The recursive bisection algorithm naturally partitions the state along this east-west divide at the first level, producing a western half with approximately 83 districts and an eastern half with 15 districts.

Results. Mean PP: 0.412 (above the 50-state average of 0.401 for block-group resolution). Maximum population deviation: 0.31%. County splits: 11 (vs. 14 in the enacted 2021 map). Runtime: 180 seconds (block-group resolution).

The high compactness reflects Washington's relatively uniform population distribution in the western half: Puget Sound-area districts can be drawn as compact near-squares around cities and suburbs without the elongated coastal configurations that reduce compactness in eastern and southern states.

5.2 Texas: 150 Districts, Tract Resolution

Texas's House of Representatives has 150 districts ($k = 150$) with a 2020 population of 29,317,186, yielding $T^* = 195{,}448$. Texas has 5,265 census tracts ($k/n = 0.028$), below the threshold for block-group resolution.

The geographic challenge in Texas is extreme: four major metropolitan areas (Houston, Dallas-Fort Worth, San Antonio, Austin) contain over 70% of the population in roughly 15% of the state's land area. The algorithm must draw a large number of compact urban districts in dense areas while also covering the vast, sparsely populated West Texas and Panhandle regions with geographically large but compact districts.

Results. Mean PP: 0.374 (below the 50-state average, reflecting Texas's challenging urban-rural geography). Maximum population deviation: 0.44%. County splits: 48 (vs. 62 in the enacted 2021 map). Runtime: 95 seconds.

Texas illustrates a general pattern: states with extreme population concentration in a few metropolitan areas achieve lower average compactness because urban districts must be carved into compact but geometrically constrained shapes. The 23% reduction in county splits (48 vs. 62) is consistent with the algorithm's preference for geographic coherence over political subdivision boundaries.

5.3 New Hampshire: 400 Districts, Block-Group Resolution

New Hampshire's House of Representatives is the largest state legislative chamber in the U.S. and one of the largest deliberative bodies in the world, with 400 members. With a 2020 population of 1,377,529, the ideal district population is $T^* = 3{,}444$—roughly the size of a small town. New Hampshire has only 321 census tracts, making redistricting at tract resolution impossible ($k/n = 1.25 > 1$). We use block-group resolution: New Hampshire has 698 block groups, giving a ratio of $k/n_{\mathrm{bg}} = 0.57$. This ratio is still above the optimal 20-units-per-district threshold, but block-group redistricting is feasible and produces valid non-empty districts.

Results. Mean PP: 0.388. Maximum population deviation: 0.48%. County splits: 31 (New Hampshire has 10 counties, so this represents a moderate level of splitting relative to the large number of districts). Runtime: 420 seconds.

New Hampshire illustrates the hardest case for the pipeline. With $k/n_{\mathrm{bg}} = 0.57$, the average block group contains only 1.75 districts—meaning many districts must consist of a single block group or split block groups. The pipeline handles this correctly via population-weighted sub-unit assignment when a block group is the only unit available to complete a district.

5.4 Nebraska: Unicameral Legislature, 49 Districts

Nebraska is the only unicameral state legislature in the U.S. The Nebraska Legislature has 49 members (called senators), with a 2020 population of 1,961,504 and ideal district population $T^* = 40{,}031$. Nebraska has 570 census tracts ($k/n = 0.086 > 0.05$), requiring block-group resolution (1,231 block groups, $k/n_{\mathrm{bg}} = 0.040$).

Results. Mean PP: 0.421 (highest among the block-group-resolution states). Maximum population deviation: 0.38%. County splits: 8. Runtime: 165 seconds.

Nebraska's high compactness reflects its geography: the state is roughly rectangular, population is distributed along the Platte River corridor from Omaha to Lincoln westward, and the Sandhills region occupies a large but very sparsely populated central area. The algorithm naturally groups Omaha-area block groups into compact eastern districts and distributes the few population centres of western Nebraska into geographically large but compactly bounded districts.

Because Nebraska has no upper chamber, the NestSection algorithm (F.2) does not apply. This simplifies the problem relative to other states: there is no constraint to maintain senate-house nesting.

6 Comparison to Enacted Maps

We compare algorithmic maps to enacted 2020-cycle state house maps for the 38 states that had enacted their new maps by the time of this analysis. The comparison focuses on three metrics: compactness (Polsby-Popper), population balance, and county splits.

6.1 Compactness Comparison

The algorithmic maps are more compact (higher mean PP) than enacted maps in 34 of 38 states (89%). The mean advantage is +0.081 PP points (algorithmic $\overline{PP} = 0.381$ vs. enacted $\overline{PP} = 0.300$ for the same 38 states).

The four states where enacted maps are more compact than algorithmic maps (Vermont, Hawaii, Alaska, and Wyoming) are all small, geographically compact states where the enacted maps are essentially forced to be compact by the state's shape and population distribution. In these states, the enacted map and the algorithmic map are nearly identical, and the small PP differences are within the noise of the metric.

6.2 Population Balance Comparison

All 50 algorithmic maps achieve $\leq 0.5\%$ maximum deviation. Of the 38 enacted maps, 26 (68%) achieve $\leq 0.5\%$; 12 (32%) have maximum deviations in the 0.5–5.0% range permitted by *Reynolds v. Sims*. No enacted map exceeds 5%.

The algorithmic maps are thus more population-balanced than enacted maps by a statistically significant margin: a paired t-test of maximum deviations gives $t = 3.8$, $p < 0.001$.

6.3 County Splits Comparison

As reported in Section 4.3, the algorithmic maps split counties fewer times than enacted maps in 30 of 38 states (79%). The mean reduction of 18% is statistically significant ($t = 4.2$, $p < 0.001$).

The magnitude of the reduction varies dramatically across states. States with historically aggressive gerrymandering (Pennsylvania, Wisconsin, Ohio, North Carolina) show the largest reductions (30–41%), while states with citizen redistricting commissions (California, Colorado, Michigan) show smaller reductions (5–12%). This pattern is consistent with the hypothesis that enacted maps in gerrymander-prone states deliberately crack communities across county lines, and that the algorithmic maps—which have no partisan inputs—avoid such cracking.

6.4 National Summary

7 Limitations

Block-group resolution ceiling. For the five states with $k/n_{\text{bg}} > 0.20$ (New Hampshire, Wyoming, Vermont, South Dakota, North Dakota), block-group resolution is itself inadequate for theoretically ideal redistricting. Full compliance with the 20-units-per-district

Metric	Algorithmic	Enacted
Mean PP (38 states)	0.381	0.300
States more compact than enacted	34/38	—
Max population deviation $\leq 0.5\%$	50/50	26/38
Mean county splits	22.4	27.4
States with fewer splits than enacted	30/38	—

Table 5: Aggregate comparison of algorithmic vs. enacted state house maps (38 states with available enacted 2020-cycle maps). The algorithmic maps are more compact, more population-balanced, and split fewer counties in the large majority of states.

heuristic would require block-level redistricting, which we defer due to the ~ 40 GB data requirement and $\sim 10\times$ longer runtimes. Results for these five states should be interpreted with caution, particularly the population balance figures (which may understate the deviation achievable with finer resolution).

Single-seed results. All 50-state results use seed 42. Companion paper B.7 [Della-Libera, 2026h] shows that for congressional redistricting, seed sensitivity is primarily a concern for states with $k \geq 8$ at the congressional scale. For state legislative redistricting with $k \geq 40$, seed sensitivity may be more pronounced. We report 5-seed sensitivity checks for the four case study states; all produce identical partisan outcomes and PP scores within ± 0.005.

No political data. As with all `redist` pipeline outputs, the state house maps in this paper use no political data: no precinct-level election returns, no partisan registration figures, no racial data. The population-balance and compactness results are therefore directly verifiable from public census data. Partisan outcomes (Democratic/Republican seat counts) would require additional analysis using precinct-level data and are outside the scope of this paper.

VRA compliance not assessed. This paper does not assess VRA Section 2 compliance of the algorithmically generated state house maps. Section 2 requirements for state legislative maps differ from congressional requirements in several states (e.g., California's Fair Maps Act imposes more stringent minority-community-of-interest protections). VRA compliance is addressed in companion paper F.6 [Della-Libera, 2026j].

Comparison data availability. Enacted 2020-cycle state house maps were available for 38 states at the time of analysis. 12 states had not yet finalized their maps or did not release machine-readable shapefiles. Results for these 12 states (county-splits comparison only) will be updated as data becomes available.

8 Conclusion

We have presented the first systematic 50-state empirical study of algorithmically generated state house maps using the minimum-edge-cut recursive bisection algorithm. The results establish three main findings.

First, the `redist` pipeline scales gracefully to the state legislative context. All 50 state house chambers are successfully redistricted, with population balance within $\pm 0.5\%$ in every state and runtimes under 8 minutes per state at single-seed operation.

Second, algorithmic state house maps are more compact than both congressional algorithmic maps ($\overline{PP} = 0.381$–0.401 vs. 0.361) and enacted state house maps ($\overline{PP} = 0.381$ vs. 0.300 for the 38-state comparison). This counterintuitive result is explained by scale effects (F.5): smaller districts have relatively shorter perimeters when drawn from compact sub-regions of the state, and recursive bisection naturally exploits this by hierarchically partitioning the state from large to small geographic scales.

Third, the algorithmic maps split counties 18% fewer times on average than enacted maps, consistent with the algorithm's lack of partisan incentives to crack geographic communities across administrative boundaries.

These results position the F track as a credible algorithmic baseline for state legislative redistricting reform. A state redistricting commission seeking a compact, population-balanced, county-respecting benchmark plan can generate one using `redist state -chamber house -resolution block_group` in under 10 minutes per state.

Future work. Immediate extensions include: partisan outcome analysis using precinct-level data (requires the political data analysis pipeline, out of scope here); VRA compliance assessment (F.6); bicameral nesting (F.2); and resolution refinement for the five states where block-group resolution is inadequate (F.3). A longer-term goal is a multi-seed ensemble analysis for the high-k chambers, building on the ensemble methodology of the G track [Della-Libera, 2026c].

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